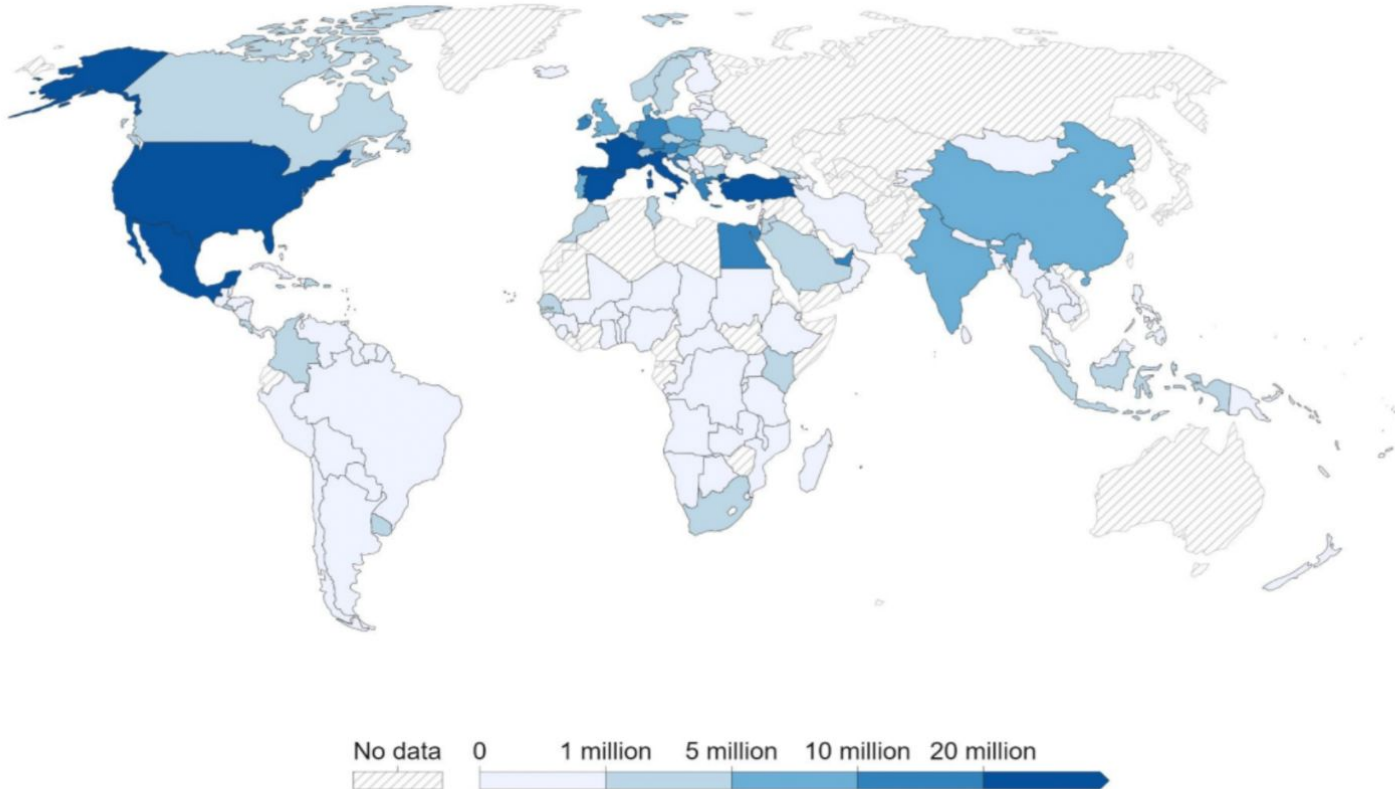


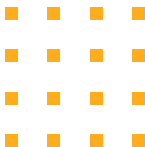
What do you see? think? wonder?

International tourist trips, 2021

Trips by people who arrive from abroad and stay overnight.

Our World
in Data





1.1 Introduction to Maps



Objective and Essential Learning

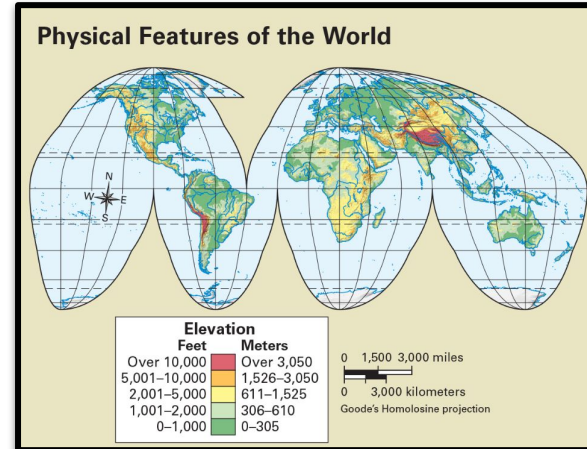
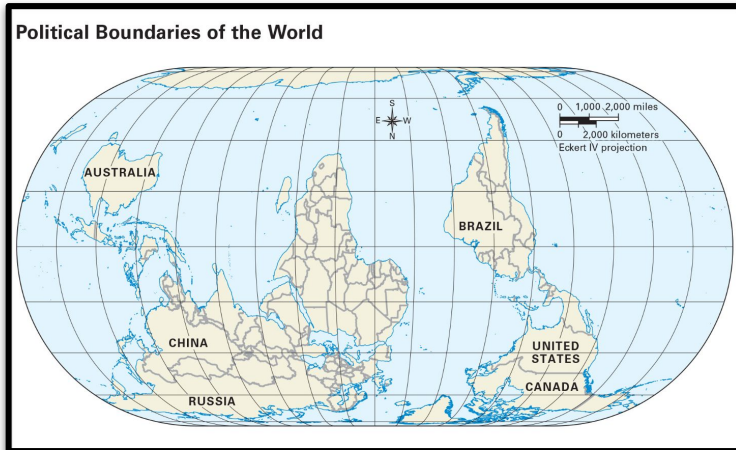
1.1 Identify types of maps, the types of information presented in maps, and different kinds of spatial patterns and relationships portrayed in maps.



- 1.1.1 Types of maps include reference maps and thematic maps.
- 1.1.2 Types of spatial patterns represented on maps include absolute and relative distance and direction, clustering, dispersal, and elevation.
- 1.1.3 All maps are selective in information; map projections inevitably distort spatial relationships in shape, area, distance, and direction.

1.1.1 Types of Maps

1. Reference Maps - general information/navigation/location
 - a. Political - states/countries/capitals
 - b. Physical- natural features
 - c. Road- highways, streets, etc.



1.1.1 Types of Maps

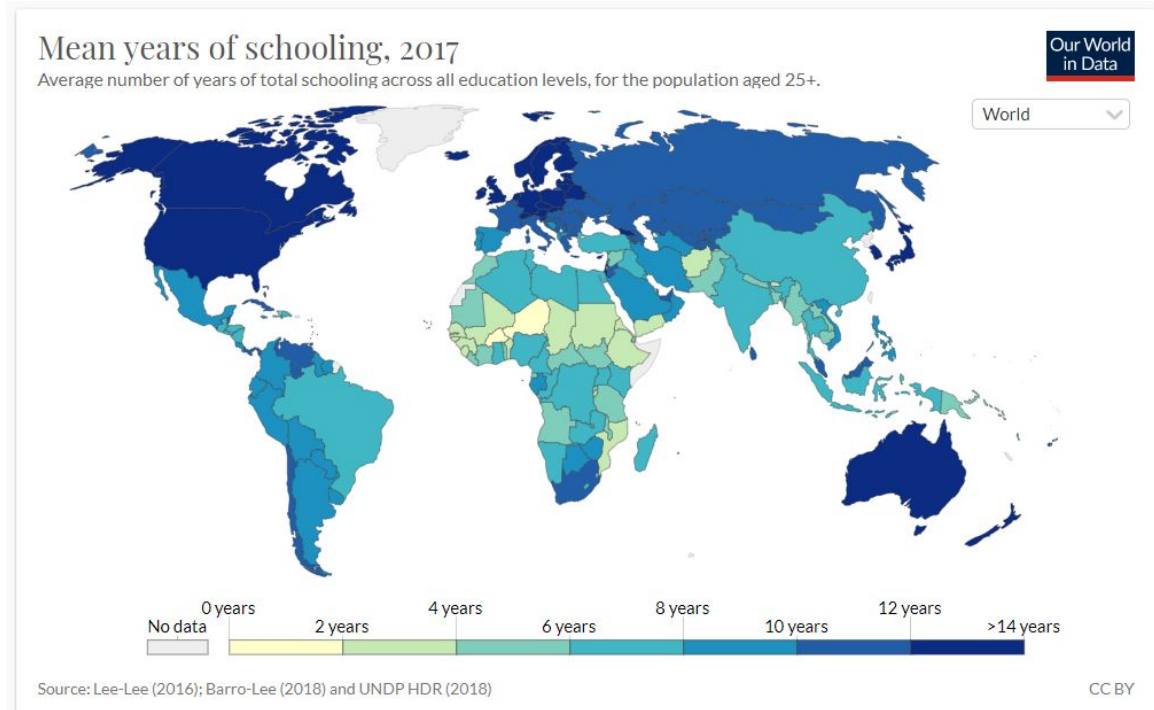
2. Thematic Maps- communicate information about a place - spatial aspects - what is it like there?

- a. Choropleth
- b. Dot-Density
- c. Graduated/Proportional Symbol
- d. Cartogram
- e. Isoline & Topographic

1.1.1 Types of Maps

Choropleth Maps

Use various colors, shades of one color, or patterns to show the location and distribution of spatial data.



1.1.1 Types of Maps

Dot-Density

Each dot represents a specified quantity of a spatial characteristic.

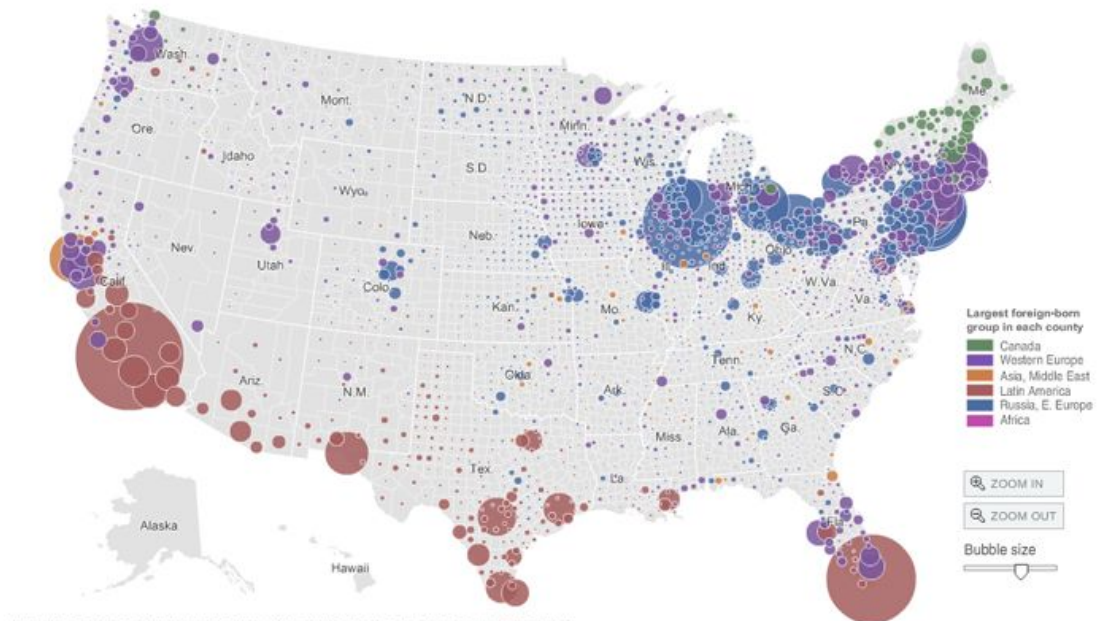


Gun Violence in Cincinnati, Ohio

1.1.1 Types of Maps

Graduated/ Proportional Symbol

Use symbols of different sizes to indicate different amounts of a variable.



Note: Due to limitations in the Census data, foreign-born populations are not available in all areas for all years.

Sources: Social Explorer, www.socialexplorer.com; Minnesota Population Center; U.S. Census Bureau

Matthew Bloch and Robert Gebeloff/The New York Times

1.1.1 Types of Maps

Cartogram

The sizes of countries are shown according to a specific variable. Area is distorted to show a variable.



World Population in 2018

The country's size in this map represents the size of the population. Each square [■] represents 500,000 people. All 15,266 squares show where the world's 7.633 billion people live.

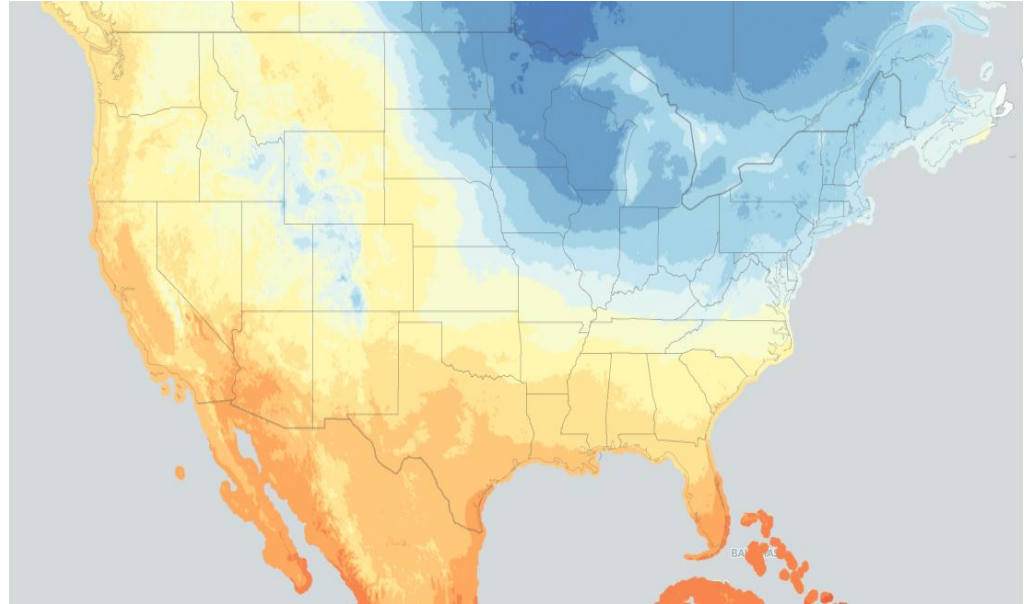
by Max Roser for Our World in Data.org – the free online publication that presents the data and research on how the world is changing.
Population data from the UN Population Division, Version 1 (September 2018). Licensed under CC-BY-SA



1.1.1 Types of Maps

Isoline

Use lines that connect points of equal value to depict variations in the data across space. Used for weather and elevation.



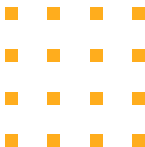
Temperature on January 31st, 2019

Topographic Maps

Elevation shown
through lines & their
distance apart.

Elevation shown
through textured
areas of map.





1.1 Introduction to Maps



Objective and Essential Learning

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1.1.2 Types Spatial Patterns on Maps

1. Absolute Location

- EXACT, PRECISE
- Address
- Latitude & Longitude
- Rio de Janeiro is located at 23°S, 43°W.

2. Relative Location

- Relationship to another place
- Next to my house
- Between Werk & Lawrence

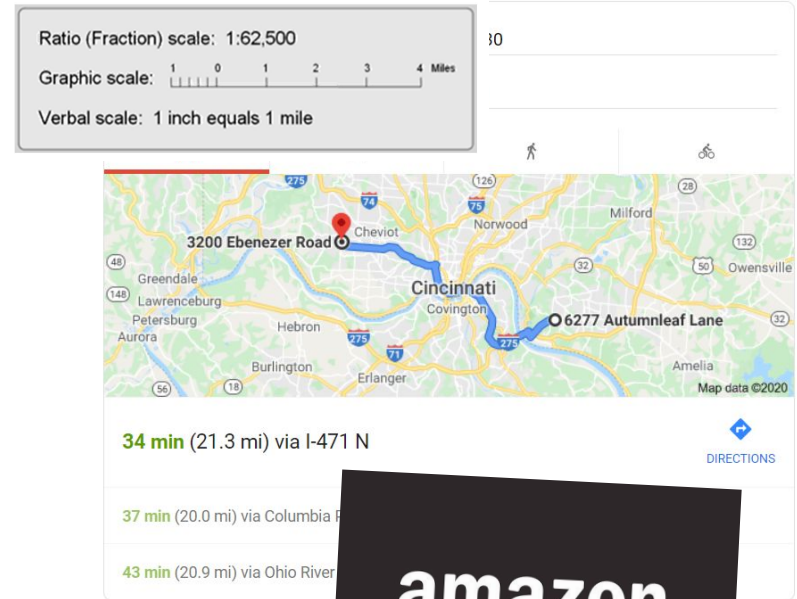
Cities Around the World



1.1.2 Types Spatial Patterns on Maps

3. Absolute Distance

- EXACT, PRECISE
- Miles/Kilometers/Feet
- LBHS is 15.3 miles away from my house.
- Map Scale



4. Relative Distance

- Spatial Interaction: Connections, contacts, movement, and flow of things between places.
- Amazon has drastically decreased relative distance between consumers and products.
 - I can order anything on Prime and have it delivered to my house in two days!



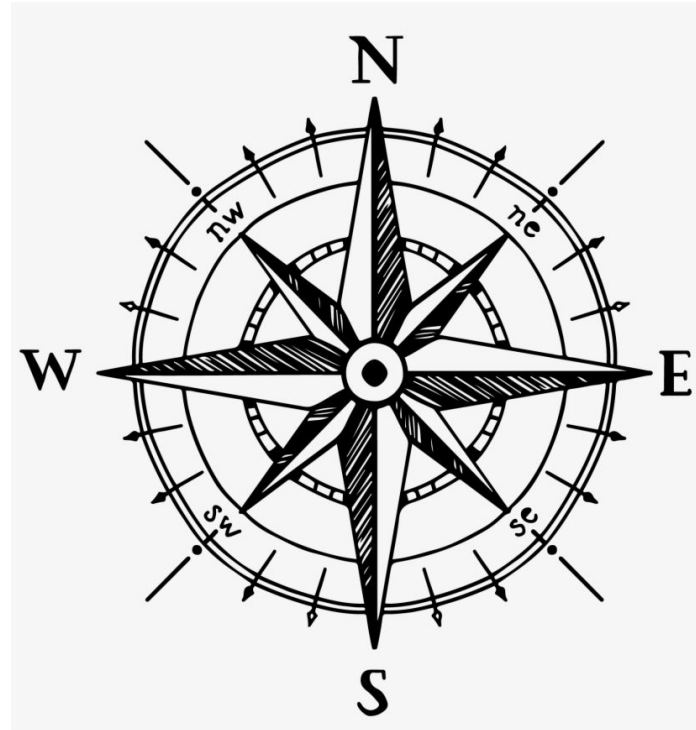
1.1.2 Types Spatial Patterns on Maps

5. Absolute Direction

- EXACT, PRECISE
- Cardinal Directions: NSEW

6. Relative Direction

- Remember, relative-relationships.
- Turn left at the Target, then right at the second stop-sign



1.1.2 Types Spatial Patterns on Maps

7. Clustering

- Close together
- Density - The # of something in a defined area.

8. Dispersal/Distribution

- Far apart
- Distribution - The way something is spread out over an area.

9. Patterns & Spatial Associations

- Indication that two (or more) phenomena may be related, associated, or correlated with one another.

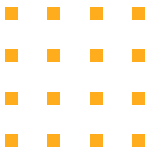
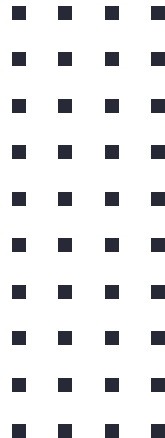


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1.1

Introduction to Maps



Objective and Essential Learning

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1.1.3 Map Projections & Distortions

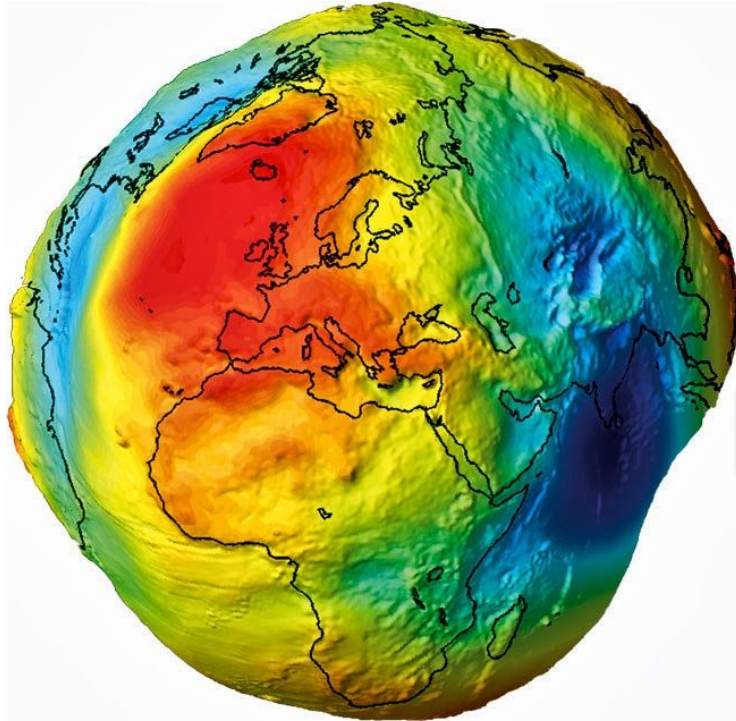
Map projections distort spatial relationships in shape, area, distance and direction.

Map Projection: The process of a cartographer (map maker) showing the curved surface of the earth on a flat surface (map).

Maps are selective in what they portray - different projections are used for different purposes.



1.1.3 Map Projections & Distortions



1.1.3 Map Projections & Distortions

Map projections are...S.A.D.D, because they distort

S- Shape

A-Area

D- Distance

D- Direction



1.1.3 Map Projections & Distortions

Mercator (1569)

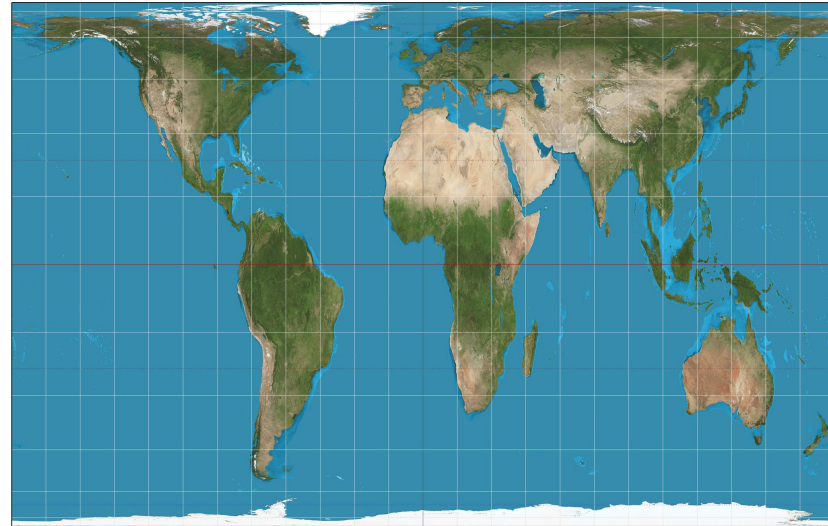
Advantages	Disadvantages
❑ Direction ❑ Shape ❑ Purpose: Navigation ❑ Preserves right angles of latitude and longitude.	❑ Area distorted near the poles ❑ Increases size of high latitude areas.



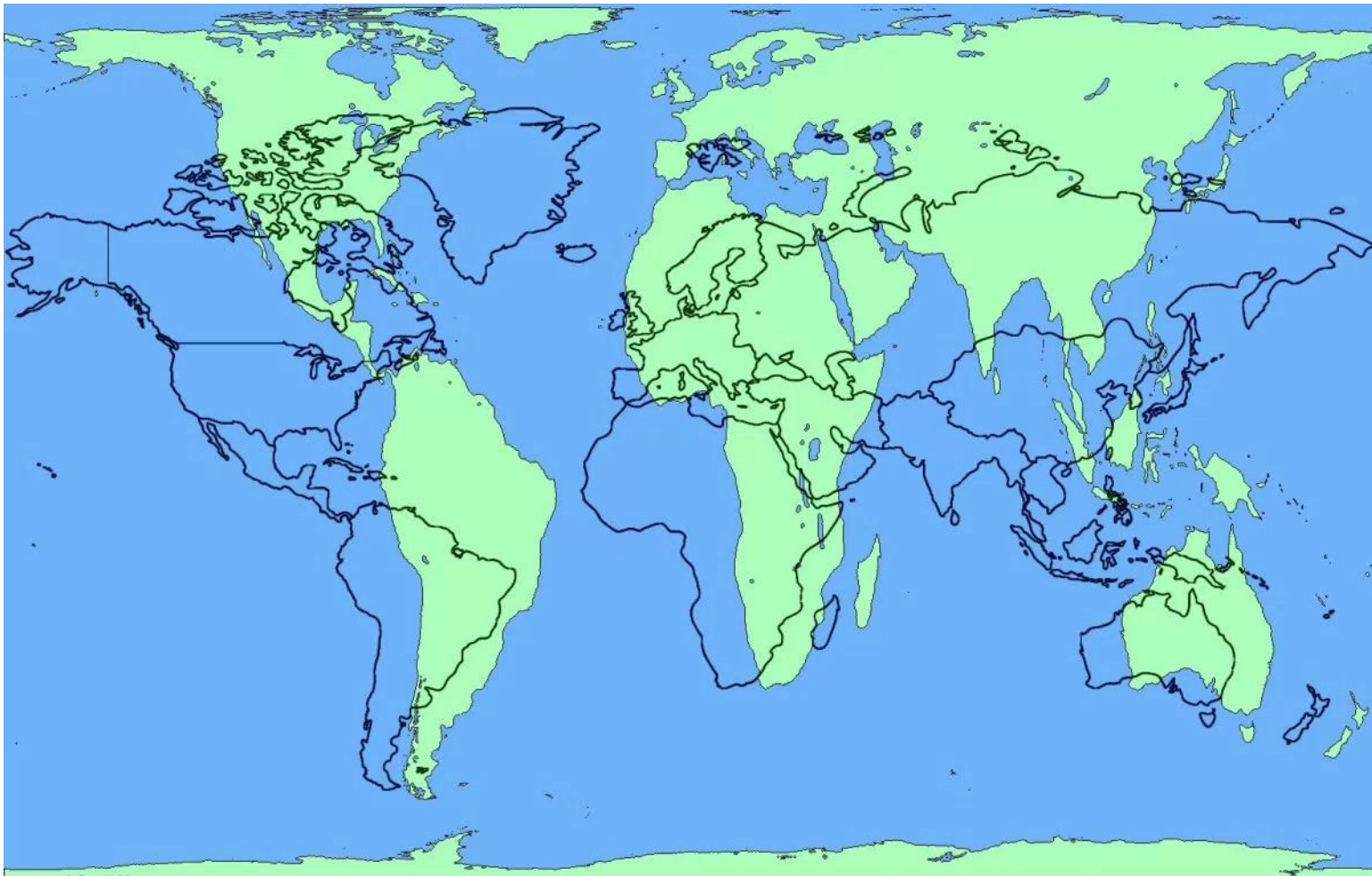
1.1.3 Map Projections & Distortions

Peters Equal Area

Advantages	Disadvantages
❑ Area of landmasses are accurate. ❑ Repositions many countries to their rightful size.	❑ Shapes are inaccurate near the poles. ❑ Vertically stretched near the equator.



What is
being
distorted
in the
green
map?



Africa VS Greenland



Mercator



Actual

What is
being
distorted
in the
Mercator
map?

Draw in
your notes

Uploaded by
@fanmaps

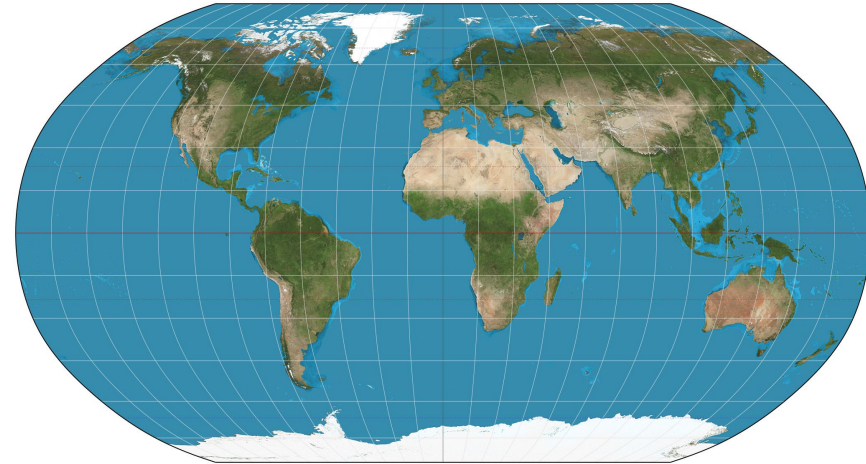


On the next page we will have a website. Please input 3 different countries to explore the real size of the country

1.1.3 Map Projections & Distortions

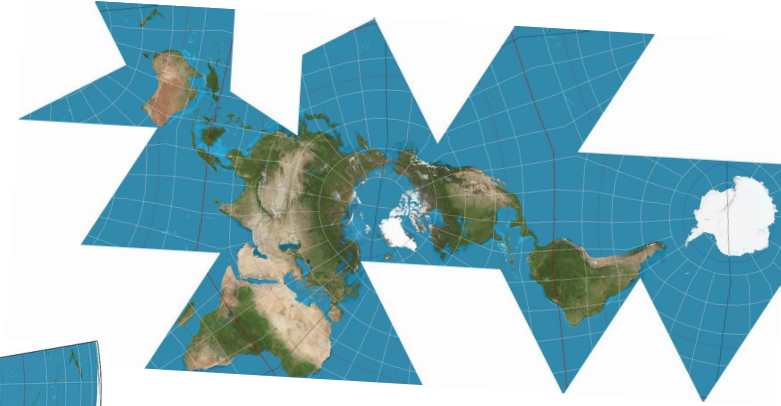
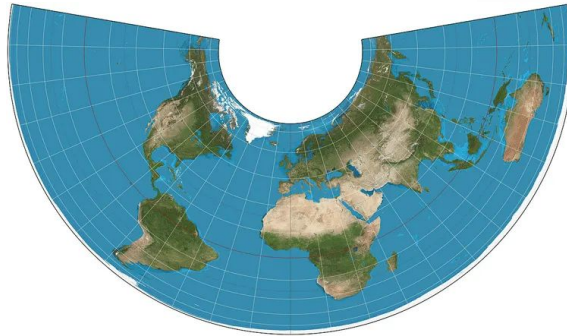
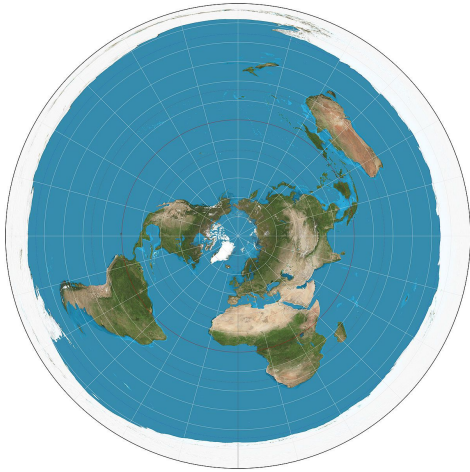
Robinson

Advantages	Disadvantages
❑ No major distortions ❑ Purpose-Compromise	❑ All aspects are slightly distorted.



1.1.3 Map Projections & Distortions

Others - Conic, Azimuthal, Fuller



1.1.3 Map Projections & Distortions

What is the point?

All maps are selective and distort the earth.

